Tomasz Młynik, PhD

Data & Software Engineer · Physicist

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PROFILE

Physicist turned data and software engineer. Lead Software Engineer at Embedded Systems DO, building Python and React applications on AWS and data pipelines for environmental and IoT monitoring (BENTOS water-quality buoy for the Gulf of Gdańsk). PhD in quantum information with a strong background in optimisation, numerical methods and reproducible analysis; I break ambiguous problems into testable pieces and explain technical things plainly.

EXPERIENCE

Lead Software Engineer · Embedded Systems DO 04/2026 – present · Gdynia

Joined as software engineer and data analyst; leading the software team since 10/2026.

- Develop and maintain web applications with a Python backend and React frontend in a large, complex codebase; build and deploy components on AWS.
- BENTOS (open-source water-quality buoy with the Institute of Oceanology PAN): data ingestion, validation and analysis workflows; dashboards, including the open-source SatBałtyk dashboard (Python/NumPy GeoTIFF pipeline, React, Vite, Leaflet).
- Lead the software team: demo readiness, test coordination and delivery priorities.

Junior Specialist, Events & Scientific Programmes · Centrum Nauki Experyment 08/2025 – 04/2026 · Gdynia

- Partnerships with universities and student organisations; science and outreach events across Poland and Europe; procurement for the unit.
- Python and scikit-learn analysis to streamline internal processes; interactive physics demos for public events (quantum lab, genetic algorithm for the travelling-salesman problem).

Administrative Assistant · Translation agency 11/2016 – 09/2018 · Gdynia

- Data-quality checks on client documents, Excel pipelines for translation workflows, invoicing.

Physics & Robotics Lecturer · Roboty i Spółka 03/2015 – 07/2016 · Gdańsk

- Taught robotics with LEGO Mindstorms EV3 and the basics of programming.

EDUCATION

PhD, Physics · University of Gdańsk 2019 – 2024

Thesis: *k-positive maps in physics*

- Quantum information: positive maps and k-entanglement witnesses, higher-order transformations of unitary operations, port-based teleportation, symmetry-reduction methods.
- Python and MATLAB code with semidefinite programming (CVX, YALMIP) to verify conjectures on finite-dimensional algebras.
- Talks: AQIS 2023 (Seoul), YITP 2023 (Kyoto). Grants: PI of UGrants-start 2 and of an academic-exchange grant (NAVA); team member of NCN Sonata 16.
- Taught algebra, stochastic processes, functional analysis and Python with algorithms.

MSc, Physics · University of Gdańsk 2015 – 2019

Thesis: *k-positive indecomposable maps on matrix algebras*

BSc, Physics · University of Gdańsk 2012 – 2015

Thesis: *Study of the galaxy's rotation curve* — exploratory data analysis and Monte Carlo simulations on satellite data of stars.

SELECTED PUBLICATIONS

F. Grosshans, M. Horodecki, M. Murao, **T. Młynik**, M. T. Quintino, M. Studziński, S. Yoshida. Multicopy quantum state teleportation with application to storage and retrieval of quantum programs. *Quantum* 10, 2105 (2026). doi:10.22331/q-2026-05-13-2105

T. Młynik, H. Osaka, M. Marciniak. Characterization of k-positive maps. *Communications in Mathematical Physics* 406, 62 (2025). doi:10.1007/s00220-025-05250-4

D. Ebler, M. Horodecki, M. Marciniak, **T. Młynik**, M. T. Quintino, M. Studziński. Optimal universal quantum circuits for unitary complex conjugation. *IEEE Trans. Inf. Theory* 69(8), 5069 (2023). doi:10.1109/tit.2023.3263771

M. Studziński, **T. Młynik**, M. Mozrzymas, M. Horodecki. Irreducible matrix representations for the walled Brauer algebra. Preprint arXiv:2501.13067 (2025).

P. Gnaciński, **T. Młynik**. Keplerian rotation of our galaxy? *Publ. Astron. Soc. Pac.* 129(974), 044101 (2017). doi:10.1088/1538-3873/aa5c9b

Full list: Google Scholar

SELECTED PROJECTS

- **BENTOS** — open-source buoy and public app for water quality and cyanobacteria risk in the Gulf of Gdańsk. Live dashboard
- **coldchain-mcp** — MCP server for local analysis of cold-chain sensor data (threshold breaches, anomalies, journey segmentation); published on PyPI. GitHub
- **Student-performance pipeline** — S3 → Lambda → Glue → Athena → Metabase with boto3, followed by gradient boosting, PCA/t-SNE/HDBSCAN and a leakage analysis.
- **Personal Finance Analyzer** — cleaning and categorising Polish bank CSV exports; CLI and Streamlit app. GitHub
- **Strange attractors & N-body simulator** — Numba-accelerated chaotic maps rendered as video; C++20/OpenGL gravity simulator with a symplectic integrator. GitHub

SKILLS

Data & Python	Python, NumPy, pandas, scikit-learn, Matplotlib, Numba, SQL, Streamlit, Excel (advanced, VBA)
Cloud & DevOps	AWS (S3, Lambda, Glue, Athena, DynamoDB), Metabase, Docker, GitHub Actions, Git
Web	React, TypeScript, Vite, Leaflet, Recharts, HTML / CSS / JS
Science & maths	Quantum information, semidefinite programming, optimisation, Monte Carlo, numerical simulation, MATLAB, SageMath, LaTeX
Learning now	C++ (OpenGL, numerics), Zephyr RTOS, TanStack Query, GraphQL
Languages	Polish (native), English (C1)